

NetMessenger – Structured Reflection

Student Details

Name: W.W.D.M. Fernando

Registration Number: IT23629844

Module: IE3010 - Network Programming

Project: NetMessenger: A Multi-Client Chat and File-Sharing Platform over TCP/IP

1. Which AI tools, if any, did you use, and at which stages of the work?

I used **ChatGPT** as a technical support and reference tool during the development of my NetMessenger application.

I used ChatGPT mainly to understand technical concepts and troubleshoot development problems. The main areas where I used AI assistance were:

- understanding the assignment requirements;
- calculating personalized project values;
- understanding TCP socket programming;
- understanding client-server communication;
- handling multiple clients;
- implementing messaging functionality;
- implementing chat rooms;
- implementing file transfer;
- understanding error handling;
- troubleshooting Git and GitHub;
- planning testing activities.

I used the AI suggestions as guidance rather than directly accepting all generated solutions. I compiled and tested the application in my own Linux environment.

2. What did the AI do well? Where did it get things wrong or mislead you?

ChatGPT was useful because it helped explain technical networking concepts in a simpler way.

It was particularly useful when I had problems understanding TCP socket programming and the relationship between the client and server.

It also helped me troubleshoot problems during development. For example, when I encountered configuration, compilation or Git-related issues, I could use AI assistance to understand possible causes and solutions.

However, I learned that AI-generated answers are not always directly suitable for my exact project.

Some suggestions needed to be changed according to my existing source code and the requirements of the assignment. Therefore, I had to test the suggestions before considering them correct.

This taught me that AI should be treated as a development assistant rather than as a replacement for testing and personal understanding.

3. What did you change, add, or reject from any AI output, and why?

I did not use every AI suggestion exactly as provided.

Before applying a suggestion, I compared it with:

- the assignment requirements;
- my existing code;
- the communication protocol;
- my project structure;
- the actual test results.

When a suggested solution did not fit my implementation, I modified it.

I also avoided unnecessary functionality that was outside the main requirements of the assignment.

The final implementation was therefore based on a combination of AI guidance, the assignment specification, my own coding decisions and actual testing.

4. What did you learn about your own understanding of network programming?

This assignment significantly improved my practical understanding of network programming.

I learned how a TCP server and client communicate using sockets.

I gained practical experience with:

- TCP connections;
- socket programming;
- client-server architecture;
- multiple client connections;
- user registration;
- message routing;
- broadcast messaging;
- private messaging;
- chat rooms;

- file transfer;
- server-side storage;
- logging;
- connection termination.

I also learned that implementing a network application requires more than simply establishing a connection. The application must also manage users, commands, messages, files and connection states correctly.

5. What was the most challenging part of the project?

One of the challenging parts was working with multiple clients.

Testing several clients at the same time helped me understand how the server needs to maintain separate information for each connected user.

File transfer was another challenging area because normal text commands and file data need to be handled correctly.

I also experienced challenges while setting up the Linux environment and Git/GitHub workflow.

Solving these problems improved my confidence in working with development environments and networking applications.

6. How did testing improve your understanding?

Testing was an important part of the project.

Instead of assuming that the application worked after compilation, I tested the actual functionality using multiple clients.

For example, I tested:

REGISTER

LIST

BCAST

PMSG

JOIN

ROOMS

RMSG

LEAVE

SENDFILE

QUIT

I also tested reconnecting clients after disconnection.

These tests helped me understand how different parts of the application interact with each other.

7. How did AI affect your learning?

AI reduced the time I needed to understand unfamiliar concepts.

Instead of only searching for individual errors, I could ask for an explanation of the underlying concept and then apply it to my code.

However, the project also showed me that understanding the answer is more important than simply copying code.

I had to compile, execute and test the implementation myself.

Therefore, I consider AI to have been a **supporting tool** rather than the main developer of the project.

8. Final Reflection

Overall, the IE3010 NetMessenger project gave me valuable practical experience in TCP/IP network programming.

I developed a better understanding of how client-server applications work and how different clients can communicate through a central server.

I also improved my ability to troubleshoot programming problems, work with Linux, use Git and GitHub, and test software systematically.

The most important lesson I learned is that **AI-generated suggestions must be validated through the assignment requirements and real testing.**

This project increased my confidence in developing network-based applications in C and gave me practical experience that I can apply to future software development projects.